

Program 2: Working with Maven: Creating a Maven Project, Understanding the POM File, Dependency Management and Plugins

Step 1: To create a Maven project run the following command in terminal:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

This command generates a basic Maven project structure with a sample App.java and AppTest.java.

```
samiksha@samiksha:~$ mkdir report
samiksha@samiksha:~$ cd report
samiksha@samiksha:~/report$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 5.8 MB/s)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] -----
[INFO] Parameter: basedir, Value: /home/samiksha/report
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/report/MyMavenApp
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 13.271 s
[INFO] Finished at: 2026-03-17T18:56:11+05:30
[INFO]
[INFO] -----
samiksha@samiksha:~/report$
```

This “BUILD SUCCESS” will confirm that the project is created successfully

Step 2: Go to the project directory using command **cd MyMavenApp**

Step 3: Open and edit pom.xml file using command **gedit pom.xml**

- Go to <https://mvnrepository.com>
- Search for a dependency “junit”
- Select the latest version and copy the dependency XML snippet.
- Paste it under the <dependencies> section in pom.xml
- Updated pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MyMavenApp</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>
<name>MyMavenApp</name>
```

```

<url>http://maven.apache.org</url>
<!-- Java version configuration -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
</properties>
<!-- Dependencies -->
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>
<!-- Build Plugins -->
<build>
<plugins>
<!-- Compiler Plugin -->
<plugin>
<groupId>org.apache.maven.plugins</groupId> <artifactId>maven-compiler-
plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>11</source>
<target>11</target>
</configuration>
</plugin>
<!-- Surefire Plugin for Testing -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<!-- JAR Plugin with Entry Point -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example.App</Main-Class>
</manifestEntries>
</archive>
</configuration>

```

```
</plugin>
</plugins>
</build>
</project>
```

Step 4: Build the project using command **mvn clean install**

```
samiksha@samiksha:~/report/MyMavenApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/samiksha/report/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---
[INFO] Building jar: /home/samiksha/report/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenApp ---
[INFO] Installing /home/samiksha/report/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/samiksha/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/samiksha/report/MyMavenApp/pom.xml to /home/samiksha/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.pom
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.566 s
[INFO] Finished at: 2026-03-17T19:08:24+05:30
[INFO] -----
```

The “Build Success” message ensures that our Maven project is built successfully.

Step 5: To run the Maven application type the command:

java -jar target/MyMavenApp-1.0-SNAPSHOT.jar

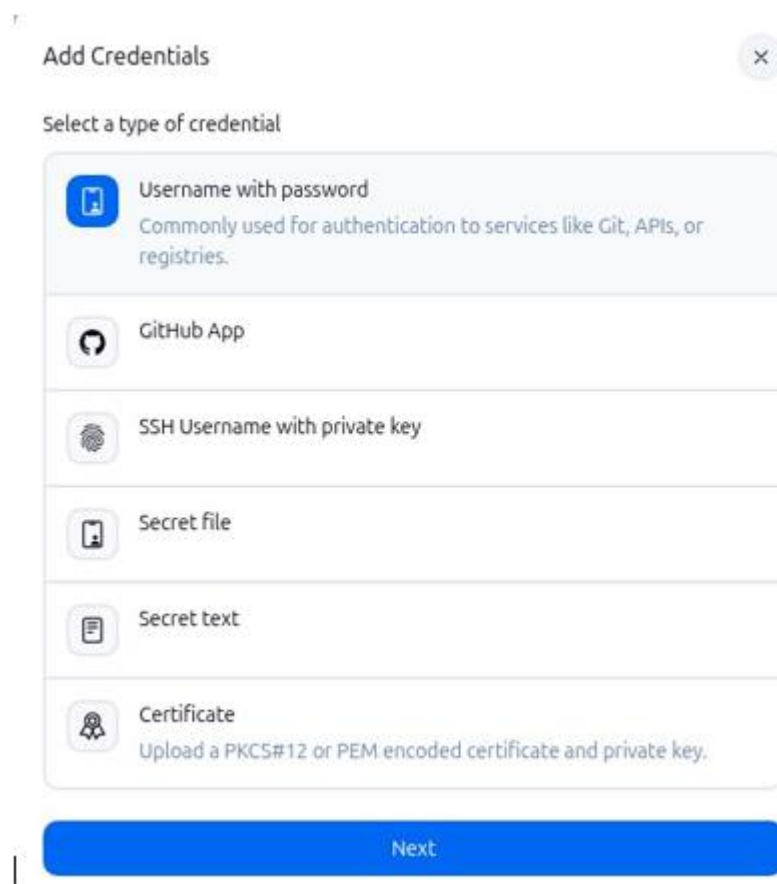
- After completing the execution, the output will look like:

```
samiksha@samiksha:~/report/MyMavenApp$ java -jar target/MyMavenApp-1.0-SNAPSHOT.jar
Hello World!
samiksha@samiksha:~/report/MyMavenApp$
```

This will show output as the code present in App.java file we can edit the text to get any other output.

Step 6: Login to Jenkins and go to the Credentials page and click on add credentials

Step 7: Add new credentials in Jenkins for CI pipeline. Enter details like username, password, set the scope to Global, and provide a description.



The screenshot shows the 'Add Credentials' dialog in Jenkins. At the top, it says 'Add Credentials' with a close button. Below that, it says 'Select a type of credential'. There is a list of six credential types, each with an icon and a description:

- Username with password**: Commonly used for authentication to services like Git, APIs, or registries.
- GitHub App**
- SSH Username with private key**
- Secret file**
- Secret text**
- Certificate**: Upload a PKCS#12 or PEM encoded certificate and private key.

At the bottom of the list is a blue button labeled 'Next'.

Step 8: now open the terminal and type the command **cd MyMavenApp** to enter the folder and create a **Jenkinsfile**. Paste the code in the Jenkinsfile

```
pipeline {
  agent any // Use any available agent
  tools {
    maven 'Maven' // Ensure this matches the name configured in Jenkins
  }
  stages {
    stage('Checkout') {
      steps {
        git branch: 'master', url: 'https://github.com/Samiksha201104/MyMavenApp.git'
      }
    }
    stage('Build') {
      steps {
        sh 'mvn clean package' // Run Maven build
      }
    }
  }
}
```

```

    }
    stage('Test') {
        steps {
            sh 'mvn test' // Run unit tests
        }
    }
    stage('Run Application') {
        steps {
            // Start the JAR application
            sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
        }
    }
}
post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}

```

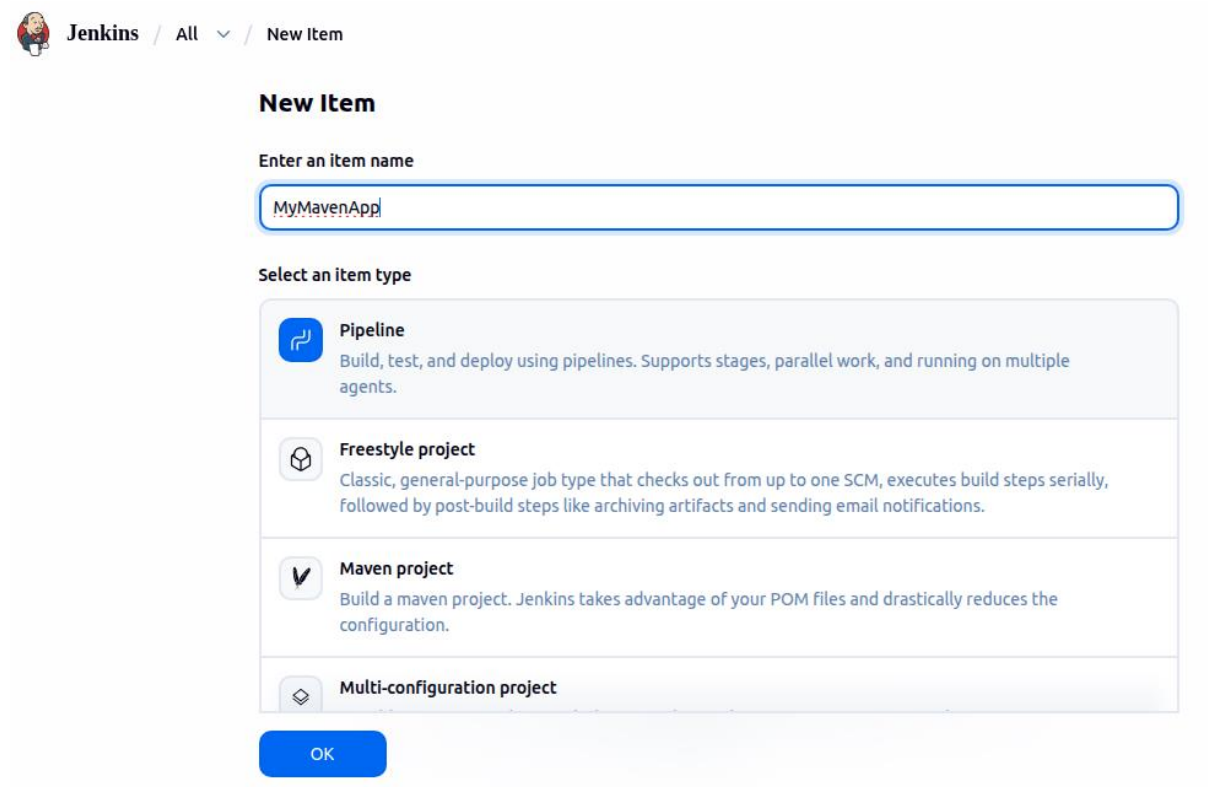
Then commit and push it to GitHub

```

samiksha@samiksha:~/Devops6D/MyMavenApp$ git init
Reinitialized existing Git repository in /home/samiksha/Devops6D/MyMavenApp/.git/
samiksha@samiksha:~/Devops6D/MyMavenApp$ git add .
samiksha@samiksha:~/Devops6D/MyMavenApp$ git commit -m "Adding Jenkinsfile"
[master 8735bc6] Adding Jenkinsfile
1 file changed, 47 insertions(+)
create mode 100644 Jenkinsfile
samiksha@samiksha:~/Devops6D/MyMavenApp$ git push -u origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 729 bytes | 729.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Samiksha201104/Devops6D.git
c5925f2..8735bc6 master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
samiksha@samiksha:~/Devops6D/MyMavenApp$ █

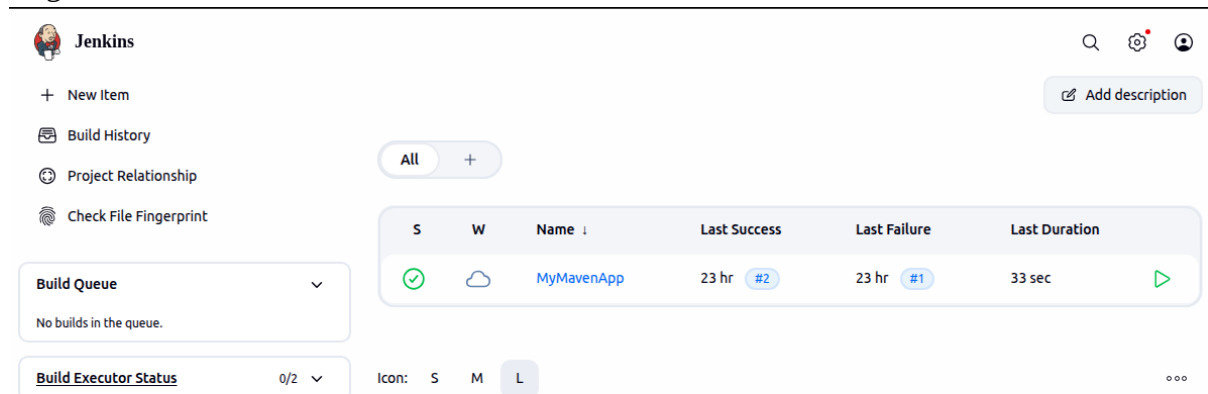
```

Step 9:Go to Jenkins Dashboard and click on new item , enter the name as MyMavenApp and choose pipeline



The image shows the 'New Item' dialog in Jenkins. At the top, there's a breadcrumb 'Jenkins / All / New Item'. The title is 'New Item'. Below it, a text field 'Enter an item name' contains 'MyMavenApp'. Underneath, a section 'Select an item type' lists four options: 'Pipeline' (with a blue icon), 'Freestyle project' (with a cube icon), 'Maven project' (with a checkmark icon), and 'Multi-configuration project' (with a folder icon). The 'Pipeline' option is selected. At the bottom is a blue 'OK' button.

Step 10: in the Jenkins dashboard select MyMavenApp and select build now and then select full stage view

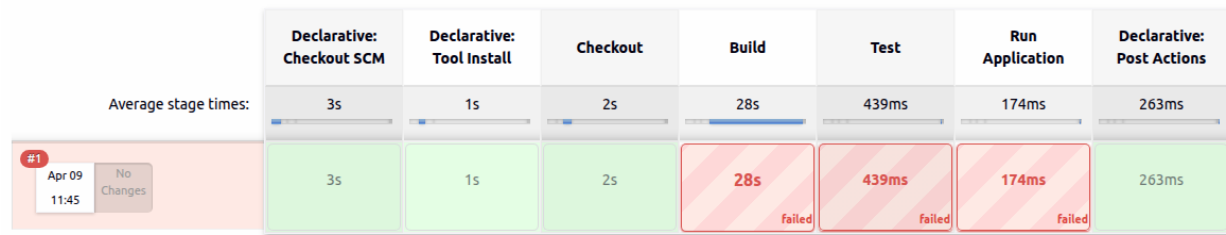


The image shows the Jenkins dashboard. On the left is a sidebar with links: '+ New Item', 'Build History', 'Project Relationship', and 'Check File Fingerprint'. Below these are two expandable sections: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2'). The main area has a search bar, a settings icon, and a user icon. Below these is a table of jobs. The table has columns: 'S', 'W', 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. There is a table with one row for 'MyMavenApp'.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	MyMavenApp	23 hr #2	23 hr #1	33 sec

Step 11 : Note the stages that failed and check the error by clicking the stage and selecting logs.

MyMavenApp - Stage View



To correct the error open pom.xml and make changes in the code

The original code has the following lines

```
<source>1.7</source>
```

```
<target>1.7</target>
```

Change this code to

```
<source>11</source>
```

```
<target>11</target>
```

Push this back to GitHub and build it again on Jenkins